Basic Info

Instructors

Teacher: Federico Bassetti (federico.bassetti@polimi.it)
TA: Bruno Ursino (bruno.ursino@polimi.i)

Scheduling

Tuesday Room: T.0.1 11:15-13:00.
Friday Room: B.5.1 14:15-16:00

Basic Info

Book:

J.M. Marin, C. Robert. **Bayesian Core: A Practical Approach to Computational Bayesian Statistics.** Springer, 2007

Exam:

The exam consists in the presentation of a statistical project. As a general rule, the projects will be developed in teams (2/3 persons) and discussed with the teacher during the course. Projects are focused on the statistical analysis of a dataset, using Bayesian tools and models. Data to analyze will be arranged together by the team and the instructor. Each students' team must hand over a project report (10-15 pages); The oral presentation of the final project is expected to be around 20/25 minutes. All the students of a group need to present some part of the projects, so please organize your presentation accordingly. For the oral presentation slides are encouraged. During and after the presentation the teachers will ask some questions, possibly to a specific student of the group. Questions can be both on the presentation and on some topic of the course (not necessarily related to the project).

tentative program

1. PART 1: bayesian learning ($\sim$ 2 (Lect) +2 (Prac. Lect.) +2 (TAlab))
   1. Principles of Bayesian learning. Likelihood, prior and posterior distributions: Bayes' Theorem. Bayesian estimation.
   2. Basic conjugate models (Beta-Bernoulli, Normal-Gamma, Poisson-Gamma).
   3. credible intervals, prediction, testing and Bayes factor.
   4. lab: basic R, data & conjugate models in practice.
2. PART 2: Monte Carlo Methods ($\sim$ 2 +1+2)
   1. Montecarlo, importance sampling, Monte Carlo Markov Chain (gibbs and metropolis)
   2. lab: Practical examples and implementation in R.
3. PART 3: Bayesian regression ($\sim$ 1 +1 +2)
   1. Linear Normal Model
   2. lab: using BAS to fit a Bayesian LM
4. PART 4: More complex models ($\sim$ 5 +4)
   1. using JAGS to fit a Bayesian model
   2. More complex models, real data & Final Project

tentative scheduling

see on we-beep

R

Download & install on your computer ”R”
starting from here
https://cran.mirror.garr.it/CRAN/
In addition to R it is useful to download and instal RStudio
https://www.rstudio.com/products/rstudio/

FOLLOW the LABS!!!

Rules of the game

- Some slides will review background material (probability), indicated as (PrRe). I will go through them quickly, but please make sure you understand the content. If something is unclear, try to fill the gaps. This is especially important for EE students, who typically have less background in probability.
- Some slides contain slightly more advanced material, labeled as (Optional). This material is not mandatory, but I encourage students who are comfortable with the basics to go through it.
- I will provide some simple exercises/homework to help you check your understanding. These are not mandatory, but I strongly recommend attempting them. They will be marked as (Hw).
- Starting from the next lecture, we will begin *hands-on* sessions using R. Additional materials, such as R scripts, R notebooks, datasets, and data sources, will be provided online for these "lab" sessions.
- **The more you engage with me, the more you will benefit from the course.**

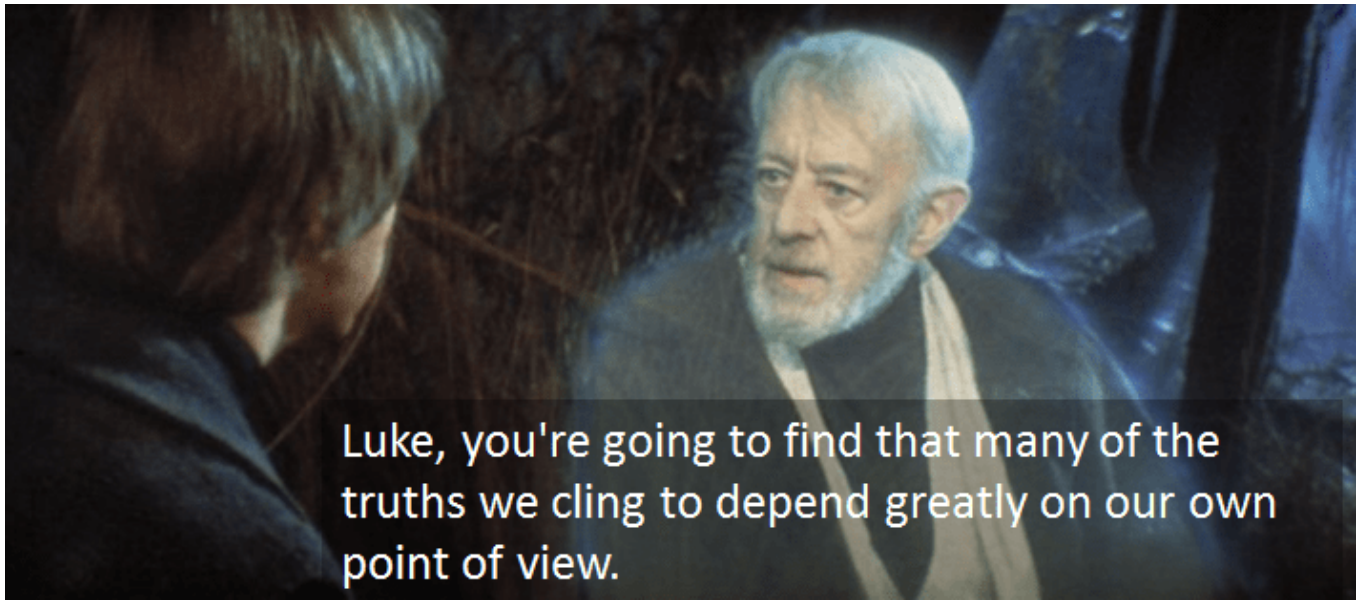
Chapter 1

the slides are not divided by lectures, but by argument: chapters

- Introduction with elementary examples: Bayes theorem.
- Review of probability: random variables and densities.
- The Bayesian core: prior, likelihood and posterior.

the bayesian side of the statistics

(hopefully not the dark one...)



[Obi-Wan Kenobi, Star Wars, Return of the Jedi]

(Almost) everything is random *(PrRe)*

the probability $P(A)$ of the event A is a number in $[0, 1]$ which quantifies the degree of belief or confidence in the truth of the event A

memories of a probability course?

- $0 \leq P(A) \leq 1$ for an event $A \subset \Omega$;
- $P(\Omega) = 1$
- $P(\cup_n A_n) = \sum_n P(A_n)$ whenever A_n are disjoint.

The Bayesian toolbox *(PrRe)*

Conditional probability

If H is such that $P(H) > 0$, the conditional probability (or posterior) of E given H is

$$P(E|H) := \frac{P(E \cap H)}{P(H)}.$$

Total probability

If $\Omega = H_1 \cup H_2$, $H_1 \cap H_2 = \emptyset$, $P(H_1) > 0$ and $P(H_2) > 0$, then

$$P(E) = P(E|H_1)P(H_1) + P(E|H_2)P(H_2).$$

hw[prove it]

Example: Circuit Diagnosis (part1)

A circuit is not working.

- Two possible **hypotheses**:
  - Dead Battery= H_1
  - Working Battery (other causes) = H_2 (note that $H_2 = H_1^c$)
- **Observation:** a voltmeter test shows "low voltage" ($= E$)
- **Initial belief/prior:** (based on your previous experience)
  - 20% chance of a dead battery, that is $P(H_1) = 0.2$
  - 80% chance of other causes, that is $P(H_2) = 0.8$
- **Model/likelihood**: $P(E|H_i)$ $i = 1, 2$ you believe that
  - $P(\text{Low Voltage}|\text{Dead Battery}) = 0.9$
  - $P(\text{Low Voltage}|\text{Working Battery}) = 0.1$

$$P(\text{Low Voltage}) \overset{tot.prob.}{=} P(E|H_1)P(H_1) + P(E|H_2)P(H_2)$$
$$= (0.9 \cdot 0.2) + (0.1 \cdot 0.8) = 0.26$$

...is this the end of the story?

The Bayesian toolbox *(PrRe)*

Bayes theorem = Inversion of probabilities

If E and H_1, H_2 are events such that $P(H_1), P(H_2), P(E) \neq 0$ and $\Omega = H_1 \cup H_2$, $H_1 \cap H_2 = \emptyset$, then

$$P(H_1|E) = \frac{P(E|H_1)P(H_1)}{P(E|H_1)P(H_1) + P(E|H_2)P(H_2)}$$

proof:

$$P(H_1|E) = \frac{P(E \cap H_1)}{P(H_1)} = \frac{P(E \cap H_1)P(E)}{P(E)P(H_1)} = \frac{P(E|H_1)P(H_1)}{P(E)}$$

$$\overset{tot.prob.}{=} \frac{P(E|H_1)P(H_1)}{P(E|H_1)P(H_1) + P(E|H_2)P(H_2)}$$

Circuit Diagnosis (part2)

- **Initial belief/prior**: $P(H_1), P(H_2)$
  - $P(\text{Dead Battery}) = 0.2$
  - $P(\text{Working Battery}) = 0.8$
- **Model/likelihood**: $P(E|H_i)$ $i = 1, 2$
  - $P(\text{Low Voltage}|\text{Dead Battery}) = 0.9$
  - $P(\text{Low Voltage}|\text{Working Battery}) = 0.1$
- **Total probability:** $P(\text{Low Voltage}) = (0.9 \cdot 0.2) + (0.1 \cdot 0.8) = 0.26$
- **Bayes' Theorem:** the posterior probability $P(H_1|E)$ is

$$P(\text{Dead Battery}|\text{Low Voltage}) = \frac{0.9 \cdot 0.2}{0.26} \approx 0.692$$

  - Prior probability: 20%.
  - Posterior probability (after the experiment): 69% .

Conclusion: The experiment significantly increases the likelihood of a dead battery.

Takeaways

Learning process: the posterior

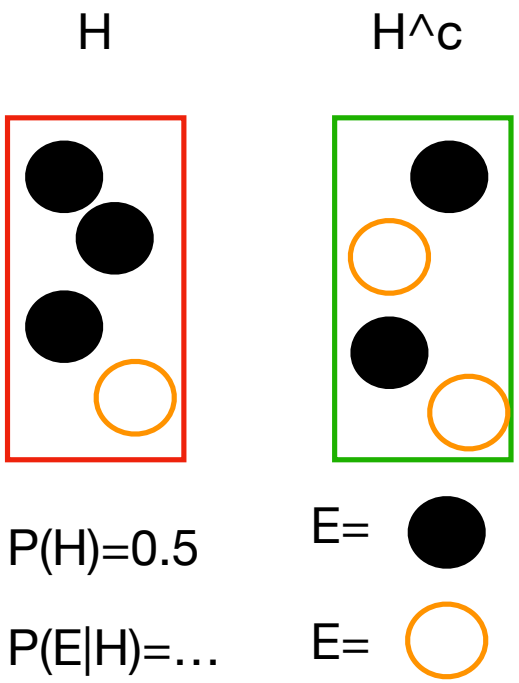
$$P(H|E)$$

is the updated probability of the hypothesis H given we observed E.

- Bayes' Theorem updates probabilities with new evidence.
- Helps make better decisions in uncertain scenarios.

Example 2: urns

- Let us consider two urns, both with four balls. In the first urn, there are 3 black balls and 1 white ball. In the second urn, 2 and 2. Randomly pick one urn. Given the urn, pick one ball.
- If the ball is black what is the (conditional) probability that it comes from urn one? (posterior probability)



$$P(\text{Urn 1}|\text{Black}) = \frac{P(\text{Black}|\text{Urn 1})P(\text{Urn 1})}{P(\text{Black}|\text{Urn 1})P(\text{Urn 1}) + P(\text{Black}|\text{Urn 2})P(\text{Urn 2})}$$

$$= \frac{\frac{3}{4}\frac{1}{2}}{\frac{3}{4}\frac{1}{2} + \frac{2}{4}\frac{1}{2}} = 3/5$$

Measures with random errors

What if the data are not $0 - 1$?

Measures with random errors.

$$X = \underset{position}{\mu} + \underset{scale}{\sigma}\ \varepsilon \qquad \text{(a.k.a. scale-position model)}$$

where

- μ is the quantity we are interested in,
- $\sigma > 0$ is the noise level, $1/\sigma$ is the precision of our measurement,
- ε random error, that is ε is a random variable.

Measures with random errors

Running Example. You are measuring the resistance R of a resistor in a circuit. Due to imperfections in measurement tools, environmental factors, and manufacturing tolerances, your measurements of R are noisy.

- You measure the resistance using a multimeter, but the multimeter is not perfectly precise and introduces its own measurement error. The measurements are approximately normally distributed with a noise level (standard deviation) of 2Ω (likelihood).
- You know from the resistor's manufacturer datasheet that the resistance R is specified as 100Ω, but it may vary slightly due to manufacturing tolerances. The data sheet states that the resistance is normally distributed with a mean of 100Ω and a standard deviation of 5Ω (prior).
- You take two measurements, and the results are $x_1 = 98\Omega$ and $x_2 = 99.2\Omega$ (observations).

$$\ldots \qquad X_i = R + 2\varepsilon_i \qquad \varepsilon_i \sim \mathcal{N}(0,1), i = 1,2 \quad \text{(likelihood)}$$

$$x_1 = 98, x_2 = 99.2 \quad \text{(data)}$$

Bayesian models

we want more!

- $0-1$ hypotheses $\rightsquigarrow$ general parameters $\vartheta \in \mathbb{R}^M$

$$H \rightsquigarrow \vartheta$$

- one observed event $\rightsquigarrow$ observation process $X \in \mathbb{R}^D$

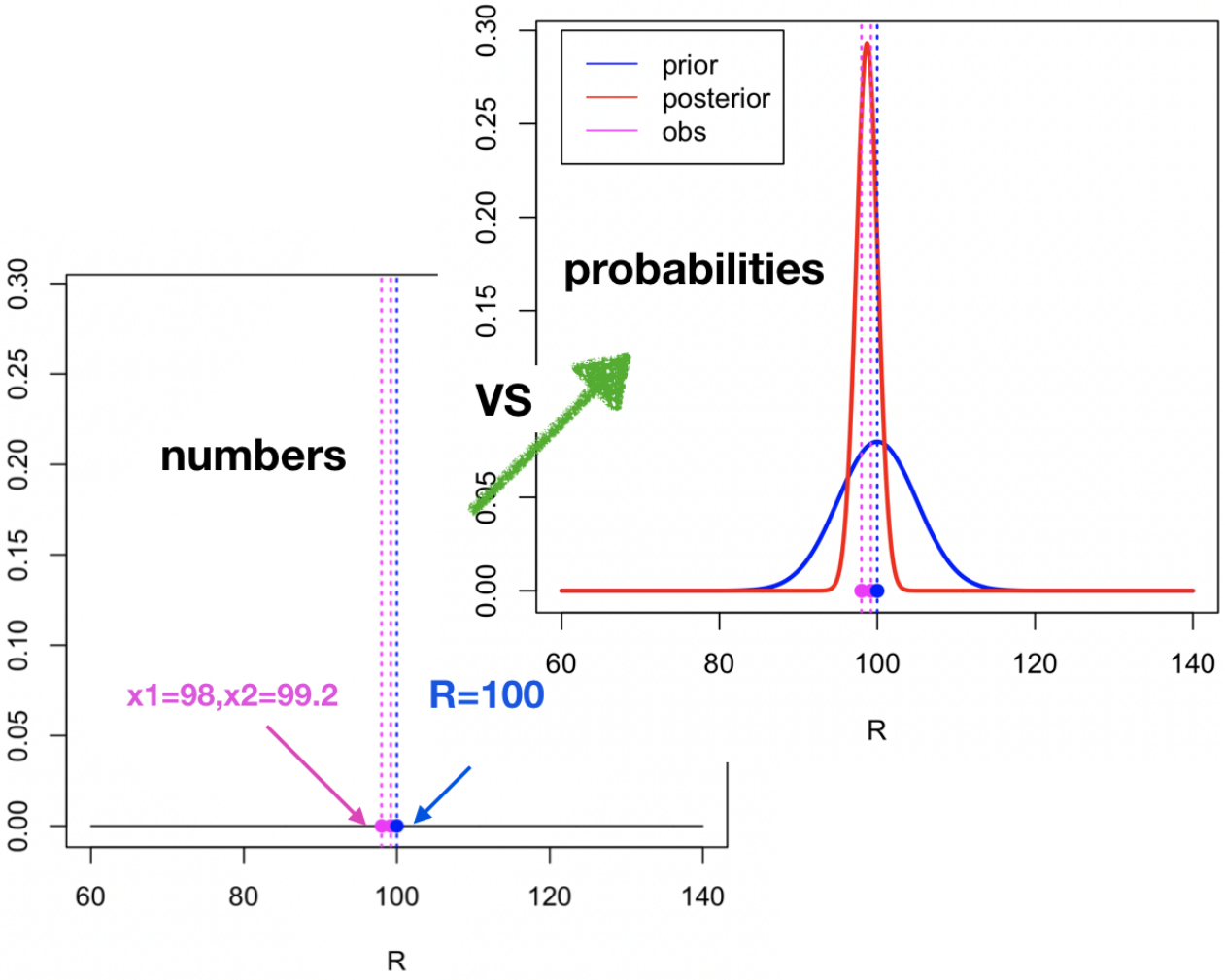
$$E \rightsquigarrow X$$

- probability of events $E, H, \ldots \rightsquigarrow$ distribution of random variables (random vectors)...

$$P(H), P(E|H) \rightsquigarrow \pi(\theta), f(x|\theta)$$

To go further we need a bit of probabilistic language.

from numbers to probabilities



Probability Review (part one)

1. Random variables and densities
2. 3 examples of statistical models
   - Bernoulli
   - Binomial
   - Normal

Random phenomena and observations (PrRe)

- A random variable (r.v.) is a number X (result of an experiment, random phenomena, etc) whose value is not completely known a priori (not deterministic)
- A random variable will be characterized by its law (probability distribution), expressed by a density $f_X(x)$. We shall consider two types of r.v.
  - Discrete r.v.
  - Continuous r.v.
- An observation is a specific realization (outcome) of X, $X = x$. We typically immagine to sample various observations from the same law (population) getting more realizations $x_1, \ldots, x_n$. Think: toss n coins!

Events and random variables *(PrRe)*

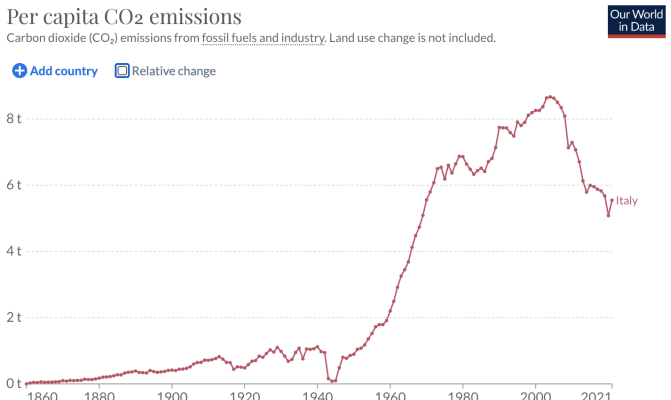
Typically one is interested in evaluating the probability of an event of the form $A = \{X \in E\}$ where $E \subset \mathbb{R}$. That is one is interested in

$$P(A) = P(X \in E).$$

Example.

X is the CO_2 emission pro capita (ton) in Italy for 2026. At the moment we don't know what will be the value of X. So it makes sense to ask what is the probability of

$$A = \{X > 5.5\}$$



Review: Random variables. *(PrRe)*

Discrete random variables

A random variable X is discrete if it takes values in a discrete (countable) set: $D = \{x_1, x_2, \dots\}$, i.e. $P\{X \in D\} = 1$. Moreover, given any set $E \subset \mathbb{R}$

$$Prob\{X \in E\} = P\{X \in E\} = \sum_{x \in E} f(x).$$

for $f(x) = p(x) = P\{X = x\}$ (probability mass function/discrete density).

Continuous random variables

The r.v. X has a continuous distribution if for every set $E \subset \mathbb{R}$

$$Prob\{X \in E\} = P\{X \in E\} = \int_E f(x)dx.$$

where the density $f : \mathbb{R} \to \mathbb{R}$ is such that $f(x) \geq 0$ and $\int_{-\infty}^{+\infty} f(x)dx = 1$.

Review: Random variables. *(PrRe)*

Bernoulli. X is a Bernoulli r.v. with parameter $\theta \in (0,1)$ if its discrete density function is

$$P\{X = x\} = f(x|\theta) = \theta^x (1-\theta)^{1-x} \text{ if } x = 0, 1$$

Clearly θ is the probability of obtaining a success. We write $X \sim \mathbf{Ber}(\theta)$.

Binomial. Let X be the number of successes that will occur in a sequence of n independent Bernoulli trials of parameter θ. The density of X turns out to be the Binomial density

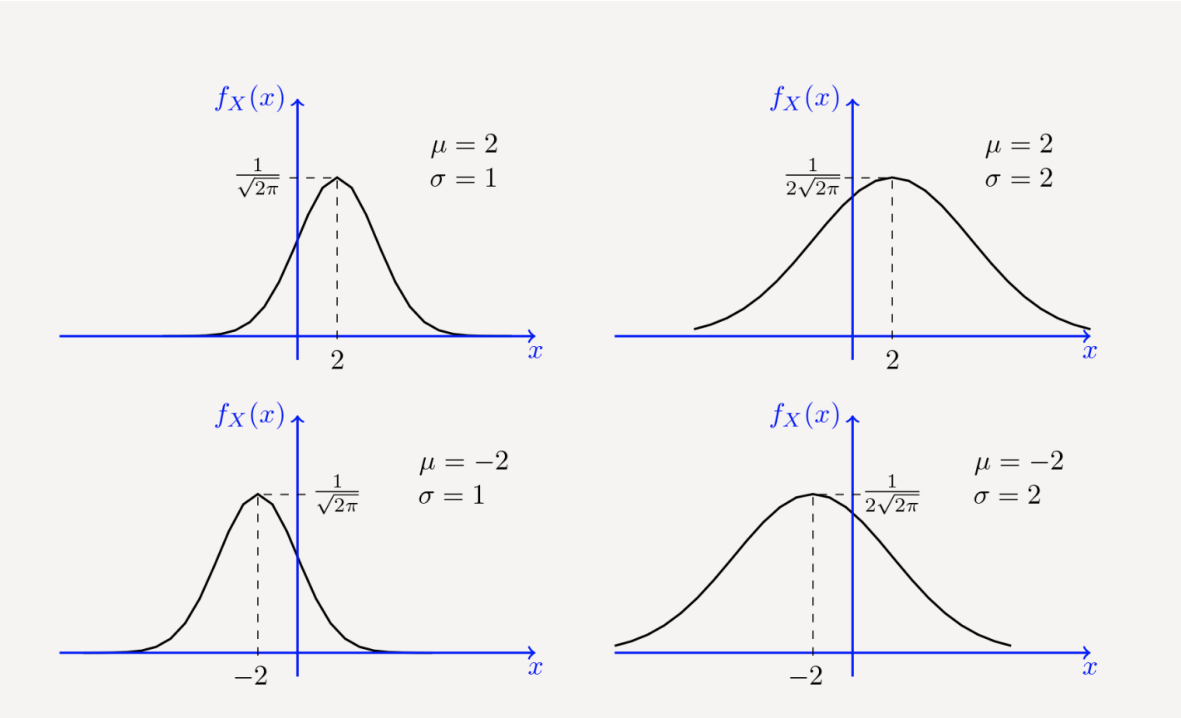
$$P\{X = x\} = f(x|\theta) = \binom{n}{x} \theta^x (1-\theta)^{n-x} \quad \text{if } x = 0, 1, \ldots, n$$

We write : $X \sim \mathbf{Bin}(n, \theta)$

Review: random variables. *(PrRe)*

Normal. X is a normal random variable of parameters (μ, σ^2), $X \sim \mathcal{N}(\mu, \sigma^2)$, if its density is

$$f(x) = \frac{e^{-\frac{(x-\mu)^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}}$$



Statistical Models *(PrRe)*

A statistical model is described by a family of densities depending on a parameter:

$$f(x|\theta): \quad x \in \mathcal{X} \subset \mathbb{R}^n \quad \theta \in \Theta$$

$x \in \mathcal{X} \subset \mathbb{R}^n$ is the variable/observation and $\theta \in \Theta \subset \mathbb{R}^m$ is the parameter

- $X \sim f(x|\theta)$ is a random variable with denisty $f(x|\theta)$ and represents the observation process,
- $X = x$ is the specific data we observe. Typically $X = (X_1, \ldots, X_n)$ is a vector!

Statistical Models: three examples

- Normal: $\mathcal{N}(\mu, \sigma^2)$:

$$f(x|\theta) = \frac{e^{-\frac{(x-\mu)^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}}, \quad x \in \mathbb{R} = \mathcal{X}, \quad \theta = (\mu, \sigma^2) \in \mathbb{R} \times \mathbb{R}_+ = \Theta$$

- Bernoulli: $\mathcal{B}er(\theta)$:

$$f(x|\theta) = \theta^x (1-\theta)^{1-x}, \quad x \in \{0, 1\} = \mathcal{X}, \quad \theta \in (0, 1) = \Theta$$

- Binomial: $Bin(n, \theta)$:

$$f(x|\theta) = \binom{n}{x} \theta^x (1-\theta)^{n-x} \quad x \in \{0, 1, \ldots, n\} = \mathcal{X}, \; \theta \in (0, 1) = \Theta$$

Probability Review (part two)

1. joint densitiy of a vector of rv
2. conditional densities
3. Bayes theorem for densities

Joint distribution. *(PrRe)*

A a random vector (X, Y) is characterized by its joint distribution and in particular by its joint density.

Joint distribution

For every set $A \subset \mathbb{R}^2$

$$P\{(X,Y) \in A\} = \begin{cases} \sum_{(x,y)\in A} f(x,y) & \text{discrete case} \\ \int_{A\subset\mathbb{R}^2} f(x,y)dxdy & \text{continuous case} \end{cases}$$

where the density $f : \mathbb{R}^2 \to \mathbb{R}$ is such that $f(x,y) \geq 0$ and $\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f(x,y)dxdy = 1$ or $\sum_x \sum_y f(x,y) = 1$.

Notation

In place of (X, Y) we shall have the vector of observations and parameters, so we typically use the symbols

$$(X, \vartheta)$$

For this reason in the next slides we directly recap some elementary fact of probability using the notation (X, ϑ) and not the commonly used (in probability) (X, Y) or (X_1, X_2) (memories of a probability course?).
Realizations of the vector (X, ϑ) will be denoted by (x, θ).

Marginal and conditional distributions. *(PrRe)*

Marginalization

If $f(x,\theta)$ is a joint density for (X,ϑ) then the marginal density of X (ϑ, respectively) is obtained by

$$f_X(x) = \begin{cases} \sum_\theta f(x,\theta) \\ \int_{-\infty}^{+\infty} f(x,\theta)d\theta \end{cases} \qquad f_\vartheta(\theta) = \begin{cases} \sum_x f(x,\theta) & \text{disc.} \\ \int_{-\infty}^{+\infty} f(x,\theta)dx & \text{cont.} \end{cases}$$

Conditional distribution

The conditional density of ϑ given $X = x$ is defined by

$$f_{\vartheta|X}(\theta|x) = \frac{f(x,\theta)}{f_X(x)}.$$

The conditional density of X given $\vartheta = \theta$ is defined by

$$f_{X|\vartheta}(x|\theta) = \frac{f(x,\theta)}{f_\vartheta(\theta)}.$$

remember that for events $P(A|B) = P(A \cap B)/P(B)$

Bayes theorem for densities (PrRe)

joint density

$$f(x,\theta) = f_{X|\vartheta}(x|\theta) f_{\vartheta}(\theta)$$

Bayes theorem

$$f_{\vartheta|X}(\theta|x) = \frac{f_{X|\vartheta}(x|\theta) f_{\vartheta}(\theta)}{f_X(x)}$$

Prior-likelihood: Bayesian models

$$H \rightsquigarrow \theta \text{ and } E \rightsquigarrow X = x$$

A Bayesian statistical model is made of

1. $f(x|\theta) = f_{X|\vartheta}(x|\theta)$: the conditional density of X given θ, called likelihood: given the parameter θ (hypotheis) the observation process X is distributed according to $f(x|\theta)$;
2. $\pi(\theta) = f_{\vartheta}(\theta)$: the marginal density of θ, called the prior: a probability distribution over the parameter space Θ.

Prior and posterior distributions

Given $f(x|\theta)$ and $\pi(\theta)$, one also has:

3. the *joint distribution* of (ϑ, X) (cf. joint density):

$$f(\theta, x) := f(x|\theta)\pi(\theta)$$

4. the *marginal distribution* of X (cf. marginalization, i.e. f_X):

$$m(x) = f_X(x) = \int f(x|\theta)\pi(\theta)\, d\theta$$

5. the *posterior distribution* of ϑ (Bayes theorem, i.e. $f_{\vartheta|X}$):

$$\pi(\theta|x) = f_{\vartheta|X}(\theta|x) = \frac{f(x|\theta)\pi(\theta)}{\int_\Theta f(x|u)\pi(u)du}$$

Bayesian Core

The posterior distribution is the center of the Bayesian inference

$$\pi(\theta|x) \propto f(x|\theta)\,\pi(\theta)$$

- Operates conditional upon the observations
- Integrate simultaneously prior information/knowledge and information brought by x
- Provides a complete inferential scope and an unique motor of inference

Frequentist and Bayesian

- Frequentist: He/she thinks of θ as an unknown but fixed parameter. Probability is only in the data, in the process of collecting data.
- Bayesian: He/she thinks that the probability is useful also in understanding the learning process, he/she treats also the parameter θ as a random variable. The learning process itself has a probabilistic nature.

Perspective

- *Uncertainty* on the parameters θ of a model modeled through a *probability* distribution π on Θ: *prior distribution*
- *Inference* based on the distribution of θ conditional on x, $\pi(\theta|x)$: *posterior distribution*
- Semantic drift from unknown θ to random θ
- Actualization of information/knowledge on θ by extracting information/knowledge on θ contained in the observation x

Bayesian learning: basic

Using

$$\pi(\theta|x) = \frac{f(x|\theta)\pi(\theta)}{\int f(x|\theta)\pi(\theta)\,d\theta}$$

you can ...

- Do Prediction
- Estimate parameters
- Derive credible regions
- Do Test
- ... and more.

Hands on: use R to understand better *(Pract. Lect.)*

Do not confuse the following different concepts

- a random variable X
- a density of a random variable X, say $x \mapsto f(x|\theta)$
- a parameter affecting a density (or describing the density of a random variable), that is the θ in $f(x|\theta)$
- a point in which we evaluate a density, e.g. if $x = 2$ and f is a standard normal density $f(2) = 0.05399$
- realization of a random variable $X = x$, say $X = 2$
- data (!)

We shall use R to visualize the previous differences in the first hand on session.

SEE THE FIRST NOTEBOOK (HandsOn_1_introR.Rmd)

Hands on: use R to understand better *(Pract. Lect.)*

- Do not make confusion between a density of a random variable (it is "a model", it is in your mind) and the histogram you get form some set of data.
- Do not make confusion between a density of a random variable and the histogram you get form simulating on a computer a sequence of (pseudo)-random number from that distribution. (we shall see what is the link between this two objects later on)

We shall use R to visualize the previous differences in the first hand on session.

SEE THE FIRST NOTEBOOK (HandsOn_1_introR.Rmd)

Chapter 2

- (conditionally) i.i.d. setting
- Bayesian normal model.
- predictive distributions
- posterior mean, posterior variance
- conjugate models

Recap from Chapter. 1

- A Bayesian model needs the likelihood of the observations $x_{1:n} = (x_1, \ldots, x_n)$, i.e. $f(x_1, \ldots, x_n|\theta)$ and a prior (pdf) for the parameter θ, i.e. $\pi(\theta)$.
- The core of the inference is the computation of the posterior distribution, that is (Bayes thm.)

$$\pi(\theta|x) = \frac{f(x|\theta)\pi(\theta)}{\int f(x|\theta)\pi(\theta)\, d\theta} \propto f(x|\theta)\pi(\theta)$$

Notation.
In probability one typically uses two different letters in order to denote a random variable (e.g. X, ϑ) and its realization (e.g. x, θ).
In what follows we will mainly use the more "statistical" notation which do not give different letters to random variables and realizations, so that we shall write x for both the random variable and its realization.
Be careful !

i.i.d. observations *(PrRe)*

The vector $(x_1, \ldots, x_n)$ is an i.i.d. samples from $f(x|\theta)$ if the joint likelihood (joint denisty of $(x_1, \ldots, x_n)$) factorizes as

$$f(x_1, \ldots, x_n|\theta) = f(x_1|\theta) \ldots f(x_{n-1}|\theta) f(x_n|\theta)$$

where $f(x_i|\theta)$ is the marginal density of the variable x_i.

i.i.d.= independent and identically distributed

An important case: i.i.d. observations

Let $x_1, \ldots, x_n$ be i.i.d. samples from $f(x_i|\theta)$ then the joint likelihood is

$$f(x_1, \ldots, x_n|\theta) = f(x_n|\theta) f(x_{n-1}|\theta) \ldots f(x_1|\theta).$$

In this case the posterior distribution of θ is

$$\pi(\theta|x_1, \ldots, x_n) = \frac{f(x_n|\theta) f(x_{n-1}|\theta) \ldots f(x_1|\theta)\pi(\theta)}{\int f(x_n|u) f(x_{n-1}|u) \ldots f(x_1|u)\pi(u)du}$$

Normal-normal model

Normal-normal

Assume that $x_1, \ldots, x_n$ are i.i.d. form a $\mathcal{N}(\mu, \sigma^2)$ (with σ^2 known) and let $\mu \sim \mathcal{N}(\mu_0, \tau^2)$ be the prior on the mean μ.

The posterior distribution is again a normal distribution $\mathcal{N}(\mu_n, \sigma_n^2)$ with

$$\mu_n = \frac{\sigma^2}{\sigma^2 + n\tau^2}\mu_0 + \frac{n\tau^2}{\sigma^2 + n\tau^2}\frac{1}{n}\sum_{i=1}^{n} x_i, \quad \sigma_n^2 = \frac{\sigma^2\tau^2}{\sigma^2 + n\tau^2}$$

Proof (optional)

The posterior distribution of μ is

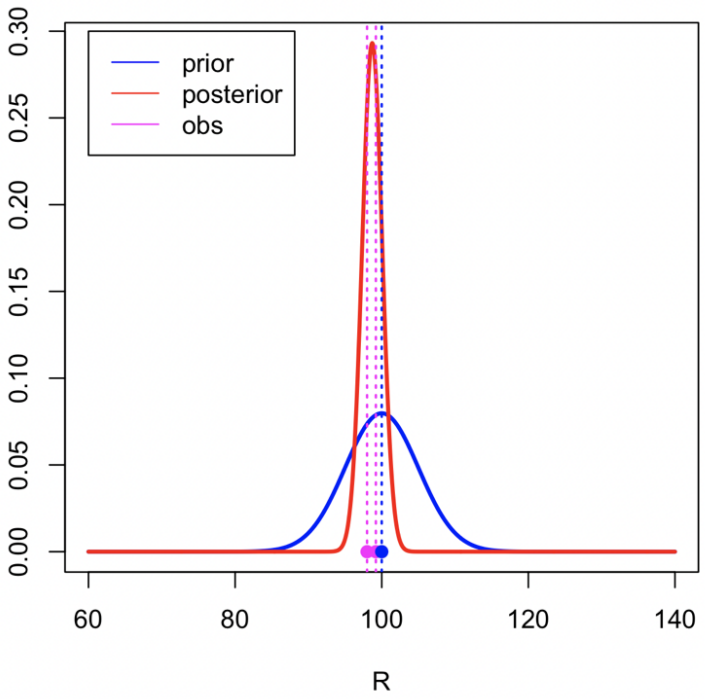
$$\begin{aligned}
\pi(\mu|x_1,\ldots,x_n) &\propto f(x_1,\ldots,x_n|\mu)\pi(\mu) \\
&\propto \exp\Big\{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2-\frac{1}{2\tau^2}(\mu-\mu_0)^2\Big\} \\
&\propto \exp\Big\{-\frac{1}{2\sigma_n^2}(\mu-\mu_n)^2\Big\}
\end{aligned}$$

with

$$\mu_n = \frac{\sigma^2}{\sigma^2+n\tau^2}\mu_0+\frac{n\tau^2}{\sigma^2+n\tau^2}\frac{1}{n}\sum_{i=1}^{n}x_i, \quad \sigma_n^2=\frac{\sigma^2\tau^2}{\sigma^2+n\tau^2}$$

back to the resistor example

- R is specified as $\mu_0 = 100\Omega$ with a standard deviation of $\tau = 5\Omega$ (prior).
- The measurements are normally distributed with a noise level $\sigma = 2\Omega$ (likelihood).
- $n = 2$ measurements: $x_1 = 98, x_2 = 99.2$ (observations).



DO THE COMPUTATIONS!

Bayesian learning: predictive distributions

The *predictive distribution* of x_{new} given x_{obs} is the conditional distribution of x_{new} given x_{obs}. This distribution depends on the joint distribution of (x_{new}, x_{obs}), which usually is given by specifying the *conditional predictive distribution* of x_{new}

$$g(x_{new}|\theta, x_{obs}),$$

that is the conditional distribution of x_{new} given the past observation(s) x_{obs} and the parameter θ.

Bayesian learning: predictive distributions

If the *conditional predictive distribution* of x_{new} given θ and x is $g(x_{new}|\theta, x)$, then the *predictive distribution* of x_{new} given x:

$$g(x_{new}|x) = \int g(x_{new}|\theta, x)\pi(\theta|x)d\theta\,.$$

predictive for cond. iid observations

Let $x_1, \ldots, x_n, x_{n+1}$ be conditionally i.i.d. given θ with density $f(x_i|\theta)$. Then for $x = (x_1, \ldots, x_n)$ and $x_{new} = x_{n+1}$ one has $g(x_{new}|\theta, x) = f(x_{new}|\theta)$ and

$$g(x_{new}|x) = g(x_{n+1}|x_1, \ldots, x_n) = \int f(x_{new}|\theta)\pi(\theta|x)d\theta.$$

proofs (optional)

The proof is as follows:

$$f(x_{new}, x, \theta) = g(x_{new}|\theta, x) f(x|\theta)\pi(\theta)$$

and hence

$$f(x_{new}, x) = \int f(x_{new}, x, \theta) d\theta = \int g(x_{new}|\theta, x) f(x|\theta)\pi(\theta) d\theta.$$

Now recall that the marginal density of x is $m(x) = \int f(x|\theta)\pi(\theta)d\theta$ and $\pi(\theta|x) = f(x|\theta)\pi(\theta)/m(x)$. Hence

$$\begin{aligned} g(x_{new}|x) &= \frac{\int g(x_{new}|\theta, x) f(x|\theta)\pi(\theta)d\theta}{\int\int g(x_{new}|\theta, x) f(x|\theta)\pi(\theta)d\theta dx_{new}} \\ &= \frac{\int g(x_{new}|\theta, x) f(x|\theta)\pi(\theta)d\theta}{\int f(x|\theta)\pi(\theta)d\theta} \\ &= \int g(x_{new}|\theta, x) \frac{f(x|\theta)\pi(\theta)d\theta}{m(x)} = \int g(x_{new}|\theta, x)\pi(\theta|x)d\theta \end{aligned}$$

Normal-normal model

Normal-normal

Assume that $x_1, \ldots, x_n$ are i.i.d. form a $\mathcal{N}(\mu, \sigma^2)$ (with σ^2 known) and let $\mu \sim \mathcal{N}(\mu_0, \tau^2)$ be the prior on the mean μ.

The posterior distribution is

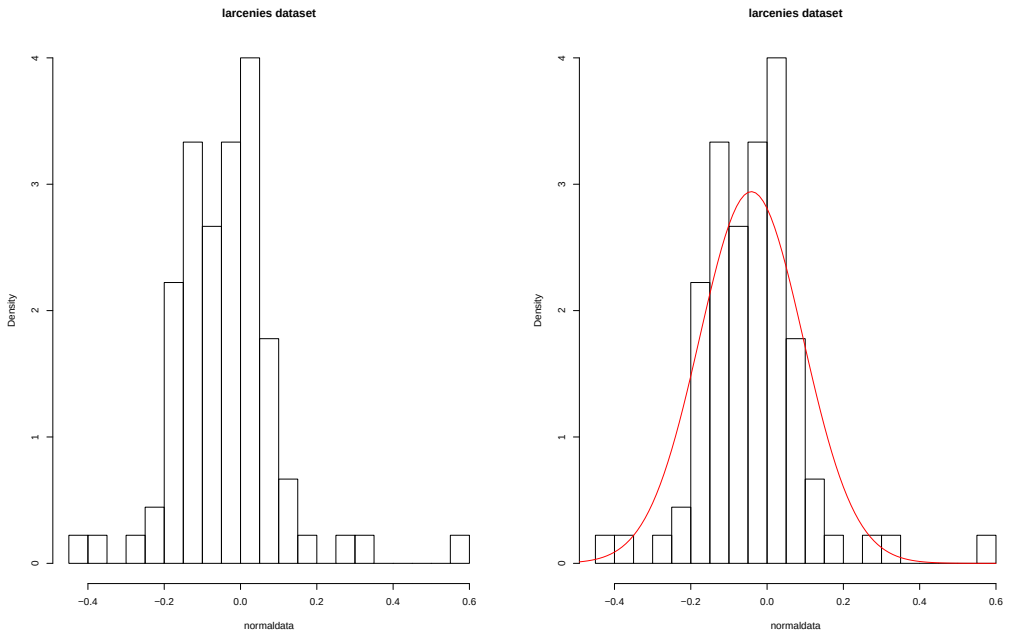
$$\mu | x_{1:n} \sim \mathcal{N}(\mu_n, \sigma_n^2) \quad \text{with}$$
$$\mu_n = \frac{\sigma^2}{\sigma^2 + n\tau^2}\mu_0 + \frac{n\tau^2}{\sigma^2 + n\tau^2}\frac{1}{n}\sum_{i=1}^n x_i, \quad \sigma_n^2 = \frac{\sigma^2\tau^2}{\sigma^2 + n\tau^2}$$

The predictive distribution is

$$x_{n+1} \sim \mathcal{N}(\mu_n, \sigma_n^2 + \sigma^2).$$

NormalData

Let $X_1, \ldots, X_n$ be the relative changes in reported larcenies between 1991 and 1995 (relative to 1991) for the 90 most populous US counties *(Source: FBI)*. What is a possible distribution for the Xs?



Download the dataset at

https://www.ceremade.dauphine.fr/~xian/BCS/normaldata

Exercise with R

1. Generate a sample of $n = 40$ observations $(x_1, \ldots, x_n)$ from a $\mathcal{N}(3, 1)$.
2. Choose prior parameters (μ_0, τ^2) (e.g. $(10, 100)$).
3. Compute the posterior distribution of μ.
4. Compute the predictive distribution of x_{41}.
5. Plot the posterior distribution.
6. Plot the predictive density against the histogram of the data.
7. Repeat the exercise with different prior parameters (e.g. $(-5, 1)$).
8. Repeat the exercise using the real data:
   https://www.ceremade.dauphine.fr/~xian/BCS/normaldata
   Assume that $\sigma^2 = 0.0183$.

Probability Review

What we need more?

1. expectation (mean) of a r.v.
2. variance of a r.v.

Expectation (Mean value) *(PrRe)*

Expectation

$$\mathbb{E}[X] := \begin{cases} \sum_x x\, f(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{+\infty} x\, f(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

provided that $\sum_x |x| f(x) < +\infty$ or t $\int_{-\infty}^{+\infty} |x| f(x) dx < +\infty$.

Thm.

Let g be a real valued function. Then

$$\mathbb{E}[g(X)] = \begin{cases} \sum_x g(x) f(x), & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{+\infty} g(x) f(x) dx, & \text{if } X \text{ is continuous} \end{cases}$$

provided $\sum_x |g(x)| f(x) < +\infty$ and $\int_{-\infty}^{+\infty} |g(x)| f(x) dx < +\infty$.

Variance. *(PrRe)*

Variance

Let $m = \mathbb{E}[X]$, then the variance of a random variable is defined as

$$Var(X) := \mathbb{E}[(X - m)^2].$$

That is

$$Var(X) = \begin{cases} \sum_x (x-m)^2 f(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{+\infty} (x-m)^2 f(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

Examples *(PrRe)*

- Normal: $X \sim \mathcal{N}(\mu, \sigma^2)$:

$$\mathbb{E}[X] = \mu, \quad Var(X) = \sigma^2$$

- Bernoulli: $X \sim \mathcal{B}er(\theta)$:

$$\mathbb{E}[X] = \theta, \quad Var(X) = \theta(1-\theta)$$

- Binomial: $Bin(n, \theta)$:

$$\mathbb{E}[X] = n\theta, \quad Var(X) = n\theta(1-\theta)$$

Posterior mean

Given the posterior distribution of θ, if θ is real valued parameter, we can compute the posterior mean (defined as the conditional mean of θ given x),

$$\mathbb{E}[\theta|x] = \begin{cases} \sum_\theta \theta\pi(\theta|x) & \text{in the discrete case} \\ \int \theta\pi(\theta|x)d\theta & \text{in the continuous case} \end{cases}$$

The posterior mean $\mathbb{E}[\theta|x]$ is used to express with a single number our "posterior" inference on a parameter θ.

Example

Let $x_1, \ldots, x_n$ be i.i.d. $\mathcal{N}(\mu, \sigma^2)$ given μ and $\mu \sim \mathcal{N}(\mu_0, \tau^2)$. Recall that the posterior distribution of μ given the data $x_{1:n}$ is the normal distribution $\mathcal{N}(\mu_n, \sigma_n^2)$ with

$$\mu_n = \frac{\sigma^2}{\sigma^2 + n\tau^2}\mu_0 + \frac{n\tau^2}{\sigma^2 + n\tau^2}\frac{1}{n}\sum_{i=1}^{n} x_i, \quad \sigma_n^2 = \frac{\sigma^2\tau^2}{\sigma^2 + n\tau^2}$$

That is

$$\mu | x_1, \ldots, x_n \sim \mathcal{N}(\mu_n, \sigma_n^2)$$

Hence we have that the posterior mean of μ is

$$\mathbb{E}[\mu | x_1, \ldots, x_n] = \mu_n.$$

Conjugate models

The prior distribution is the key to Bayesian inference

But...

In practice, it seldom occurs that the available prior information is precise enough to lead to an exact determination of the prior distribution

There is no such thing as *the* prior distribution!

Conjugate priors

The normal-normal model is an example of conjugate model...

Conjugate prior

A family $\mathcal{F}$ of probability distributions on Θ is *conjugate* for a likelihood function $f(x|\theta)$ if, for every $\pi \in \mathcal{F}$, the posterior distribution $\pi(\theta|x)$ also belongs to $\mathcal{F}$.

Only of interest when $\mathcal{F}$ is *parameterised* : switching from prior to posterior distribution is reduced to an updating of the corresponding parameters.

Justifications

- Limited-finite information conveyed by x
- Preservation of the structure of $\pi(\theta)$
- Device of virtual past observations
- Linearity of some estimators
- But mostly... tractability and simplicity

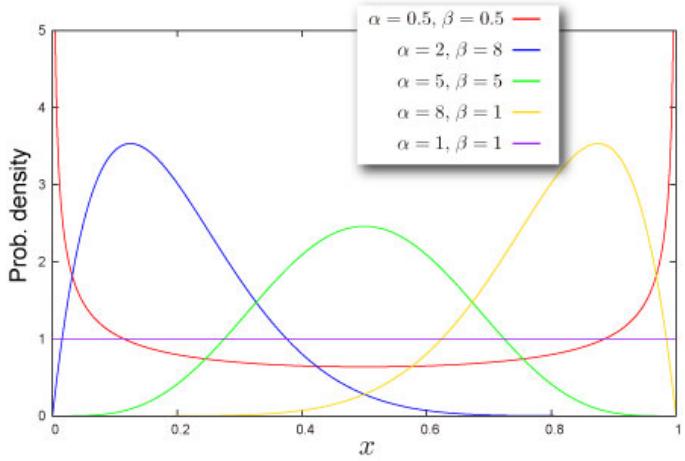
Few Important conjugate models

- Beta-Binomial Model
- Gamma-Poisson model
- Normal-Normal model
- Normal-Normal-Inverse Gamma model

Beta distribution

A generalization of the uniform (continuous) distribution on $[0,1]$ is the Beta distribution $\mathcal{B}(a,b)$, defined by the density

$$\theta \mapsto \frac{1}{B(\alpha,\beta)}\theta^{\alpha-1}(1-\theta)^{\beta-1}\mathbb{I}_{(0,1)}(\theta) \quad (\alpha>0, \beta>0)$$



$$B(a,b) := \int_0^1 x^{a-1}(1-x)^{b-1}dx = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}.$$

Binomial distribution

Number of successes

Let X be the number of successes that will occur in a sequence of n independent Bernoulli trials of parameter θ

The density of X turns out to be the Binomial density [1]

$$P\{X = x|\theta\} = f(x|\theta) = \binom{n}{x}\theta^x(1-\theta)^{n-x} \quad \text{if } x = 0, 1, \ldots, n$$

In notation: $X \sim \mathbf{Bin}(n, \theta)$

[1] note that $X = \sum_{j=1}^{n} X_j$ with X_j indep. $Ber(\theta)$

Beta-Binomial Model

Beta-Binomial Model

$$x \sim Bin(n, \theta) \quad \text{likelihood}$$
$$\theta \sim \mathcal{B}e(\alpha, \beta) \quad \text{prior}$$

That is

$$\begin{aligned} f(x|\theta) &= \binom{n}{x} \theta^x (1-\theta)^{n-x}, \qquad x = 0, 1, ..., n, \\ \pi(\theta) &= \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}, \quad 0 \leq \theta \leq 1. \end{aligned}$$

Beta-Binomial Model

The joint distribution of (x, θ) is then

$$\begin{aligned} f(x,\theta) &= \binom{n}{x}\theta^x(1-\theta)^{n-x}\frac{1}{B(\alpha,\beta)}\theta^{\alpha-1}(1-\theta)^{\beta-1} \\ &= \binom{n}{x}\frac{1}{B(\alpha,\beta)}\theta^{\alpha+x-1}(1-\theta)^{n-x+\beta-1} \end{aligned}$$

and the marginal distribution of x is

$$\begin{aligned} m(x) &= \int_0^1 \binom{n}{x}\frac{1}{B(\alpha,\beta)}\theta^{\alpha+x-1}(1-\theta)^{n-x+\beta-1}d\theta \\ &= \binom{n}{x}\frac{1}{B(\alpha,\beta)}B(\alpha+x, n-x+\beta). \end{aligned}$$

Beta-Binomial Model

Finally, the posterior distribution of θ is

$$\begin{aligned}\pi(\theta|x) &= \frac{f(x,\theta)}{m(x)} \\ &= \theta^{\alpha+x-1}(1-\theta)^{\beta+n-x-1}\frac{1}{B(\alpha+x,\beta+n-x)}\end{aligned}$$

which is a beta distribution $\mathcal{B}e(\alpha+x,\beta+n-x)$.

$$\boxed{\theta|x \sim \mathcal{B}e(\alpha+x,\beta+n-x)}$$

Beta-Binomial Model

An estimator $\hat{\theta}$ of θ is the mean of the posterior distribution:

$$\hat{\theta} = \mathbb{E}[\theta|x] = \int_0^1 \theta\pi(\theta|x)d\theta = \frac{\alpha + x}{\alpha + \beta + n}$$

Note that

$$\frac{\alpha}{\alpha + \beta} = \int_0^1 \theta \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha, \beta)} d\theta = \int_0^1 \theta\pi(\theta)d\theta = \mathbb{E}[\theta]$$

is the prior mean.

Few Important conjugate models

- Beta-Binomial Model
- Gamma-Poisson model
- Normal-Normal model
- Normal-Inverse Gamma-Normal model

Possion distribution

Number of events

Let X be the number of customers calling a help center in a day. What is a possible distribution for X?

X is a Poisson random variable of parameters $\theta > 0$ (in symbol $\mathcal{P}oi(\theta)$) if it is a discrete r.v. with density

$$f(x|\theta) = \frac{\theta^x}{x!} e^{-\theta} \qquad x = 0, 1, 2, \ldots$$

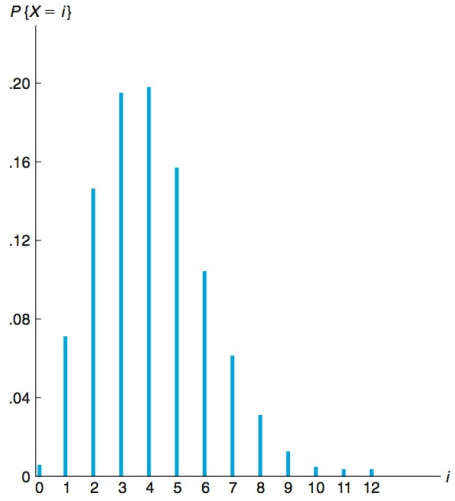


FIGURE 5.3 *The Poisson probability mass function with $\lambda = 4$.*

Gamma distribution

A r.v. λ has Gamma density $\mathcal{G}a(\alpha, \beta)$ if its density is

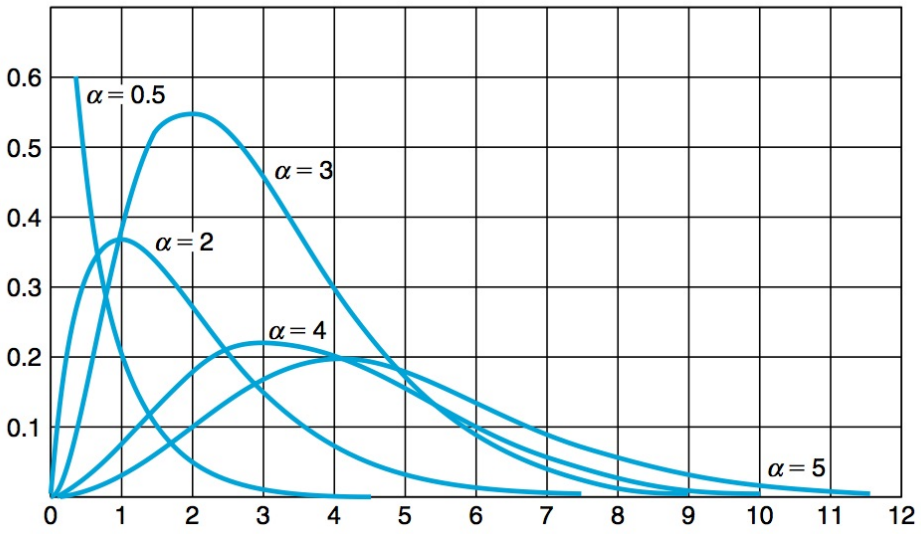
$$\lambda \mapsto \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda} \mathbb{I}\{\lambda > 0\}$$

with $\alpha > 0$ and $\beta > 0$. For $\alpha = 1$ one gets the negative exponential density (see later). Note that $1/\beta$ is a scale parameter. Be careful to the different parametrizations.

The Euler Gamma function is defined as

$$\Gamma(z) = \int_0^{+\infty} t^{z-1} e^{-t} dt$$

Gamma distribution



Above $\beta = 1$.

Gamma-Poisson model

Gamma-Poisson model

$$
\begin{aligned}
x_i &\overset{i.i.d.}{\sim} \mathcal{P}oi(\lambda) && \text{likelihood} \\
\lambda &\sim \mathcal{G}a(\alpha, \beta) && \text{prior}
\end{aligned}
$$

The likelihood function is

$$
\begin{aligned}
f(x_1, \ldots, x_n|\lambda) &= \prod_{i=1}^{n} \frac{1}{x_i!} e^{-\lambda} \lambda^{x_i} \\
&\propto e^{-n\lambda} \lambda^{\sum_{i=1}^{n} x_i}.
\end{aligned}
$$

The prior is

$$
\pi(\lambda) = \frac{1}{\Gamma(\alpha)} \beta^{\alpha} e^{-\beta\lambda} \lambda^{\alpha-1}.
$$

Gamma-Poisson model

The posterior distribution of λ is

$$\begin{aligned}
\pi(\lambda|x_1,\ldots,x_n) &\propto f(x_1,\ldots,x_n|\lambda)\pi(\lambda) \\
&\propto \left(e^{-n\lambda}\lambda^{\sum_{i=1}^n x_i}\right)\left(\frac{1}{\Gamma(\alpha)}\beta^\alpha e^{-\beta\lambda}\lambda^{\alpha-1}\right) \\
&\propto e^{-(\beta+n)\lambda}\lambda^{\alpha+\sum_{i=1}^n x_i-1}
\end{aligned}$$

The posterior is a gamma distribution $\mathcal{G}a(\alpha_n,\beta_n)$ with

$$\alpha_n = \alpha + \sum_{i=1}^n x_i, \quad \beta_n = \beta + n$$

Gamma-Poisson model

A Bayes estimator $\hat{\lambda}_n$ of λ is the mean of the posterior distribution:

$$\begin{aligned}\hat{\lambda}_n &= \int_0^{+\infty} \lambda \pi(\lambda | x_1, \ldots, x_n) d\lambda \\ &= \frac{\alpha_n}{\beta_n} = \frac{\alpha + \sum_{i=1}^n x_i}{\beta + n}\end{aligned}$$

The Bayes estimator can be written as

$$\hat{\lambda}_n = \frac{\alpha}{\beta} \frac{\beta}{n+\beta} + \hat{\lambda}_{ML,n} \frac{n}{n+\beta}$$

where α/β is the prior mean and $\hat{\lambda}_{ML,n}$ is the Maximum Likelihood (ML) estimator of λ

$$\hat{\lambda}_{ML,n} = \frac{1}{n} \sum_{i=1}^n x_i.$$

Gamma-Poisson model

The predictive distribution of x_{n+1} given $(x_1, \ldots, x_n)$ is the Negative Binomial distribution[2]

$$x_{n+1}|x_1, \ldots, x_n \sim \mathcal{NB}(\alpha_n, \beta_n(1+\beta_n)^{-1})$$

i.e.

$$g(x_{n+1}|x_1, \ldots, x_n) = \frac{\Gamma(x_{n+1}+\alpha_n)}{\Gamma(x_{n+1}+1)\Gamma(\alpha_n)} \left(\frac{\beta_n}{\beta_n+1}\right)^{\alpha_n} \left(\frac{1}{\beta_n+1}\right)^{x_{n+1}}$$

[2] $\mathcal{NB}(n, p)$: n = number of success, p probability of succes

Gamma-Poisson model

$$\mathbb{E}[\lambda|x_{1:n}] = \alpha_n/\beta_n, \quad Var(\lambda|x_{1:n}) = \alpha_n/\beta_n^2$$

while

$$\mathbb{E}[x_{n+1}|x_{1:n}] = \frac{\alpha_n}{\beta_n}, \quad Var(x_{n+1}|x_{1:n}) = \frac{\alpha_n}{\beta_n}(1 + \frac{1}{\beta_n})$$

hw[prove the results in the previous 2 slides]

Few Important conjugate models

- Beta-Binomial Model
- Gamma-Poisson model
- Normal-Normal model [already seen... just a recap!]
- Normal-Inverse Gamma Normal model

Normal-normal model

Normal-normal model

The variance σ^2 is assumed to be known and a normal prior is set on the mean of a normal sample, i.e.

$$x_i \overset{i.i.d.}{\sim} \mathcal{N}(\mu, \sigma^2) \quad \text{likelihood}$$
$$\mu \sim \mathcal{N}(\mu_0, \tau^2) \quad \text{prior}$$

One has

$$f(x_1, \ldots, x_n|\mu) \propto \exp\Big\{ -\frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2 \Big\}.$$

and

$$\pi(\mu) = \frac{1}{\sqrt{2\pi\tau^2}} \exp\Big\{ -\frac{1}{2\tau^2} (\mu - \mu_0)^2 \Big\}.$$

Normal-normal model

The posterior distribution is the normal distribution $\mathcal{N}(\mu_n, \sigma_n^2)$ with

$$\mu_n = \frac{\sigma^2}{\sigma^2 + n\tau^2}\mu_0 + \frac{n\tau^2}{\sigma^2 + n\tau^2}\frac{1}{n}\sum_{i=1}^{n} x_i, \quad \sigma_n^2 = \frac{\sigma^2\tau^2}{\sigma^2 + n\tau^2}$$

The posterior mean is

$$\mathbb{E}[\mu | x_1, \ldots, x_n] = \mu_n.$$

The predictive distribution is

$$x_{n+1} \sim \mathcal{N}(\mu_n, \sigma_n^2 + \sigma^2).$$

Full normal: Normal-Gamma prior

In the normal $\mathcal{N}(\mu, \sigma^2)$ case, when both μ and σ are unknown, there still is a conjugate prior on $\theta = (\mu, \sigma^2)$, proportional to

$$(\sigma^{-2})^{\alpha_0 - 1/2}\, e^{-\sigma^{-2}[k_0(\mu-\mu_0)^2 + 2\beta_0]/2}$$

Normal-(inverse) Gamma prior

The Normal-Normal-Gamma model is

$$\begin{aligned} x_i &\overset{i.i.d.}{\sim} \mathcal{N}(\mu, \sigma^2) \quad \text{likelihood} \\ \sigma^{-2} &\sim \mathcal{G}(\alpha_0, \text{rate} = \beta_0) \quad \text{prior} \\ \mu|\sigma^2 &\sim \mathcal{N}(\mu_0, \sigma^2/k_0) \quad \text{prior for } \mu \text{ given } \sigma^2 \end{aligned}$$

The previous prior is also known as Normal-inverse Gamma prior for (μ, σ^2), that is

$$\begin{aligned} \sigma^2 &\sim \mathcal{IG}(\alpha_0, \beta_0) \\ \mu|\sigma^2 &\sim \mathcal{N}(\mu_0, \sigma^2/k_0) \end{aligned}$$

Full normal model: Normal-Gamma prior

Setting $s_x^2 = \sum_{i=1}^n (x_i - \bar{x}_n)^2$, check that

$$f(x_1, \ldots, x_n | \mu, \sigma^2) \propto (\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2}$$
$$\propto (\sigma^2)^{-n/2}\, e^{-\left\{n(\mu - \overline{x}_n)^2 + s_x^2\right\}/2\sigma^2}$$

Then

$$\pi(\mu, \sigma^2 | x_1, \ldots, x_n) \propto (\sigma^2)^{1/2} (\sigma^2)^{-\alpha_0}\, e^{-\left\{k_0(\mu - \mu_0)^2 + 2\beta_0\right\}/2\sigma^2}$$
$$\times (\sigma^2)^{1/2} (\sigma^2)^{-n/2}\, e^{-\left\{n(\mu - \overline{x}_n)^2 + s_x^2\right\}/2\sigma^2}$$
$$\propto (\sigma^2)^{1/2} (\sigma^2)^{-\alpha_0 - n/2} \exp\left\{ -\left((k_0 + n)(\mu - \mu_n)^2 \right.\right.$$
$$\left.\left. + 2\beta_0 + s_x^2 + \frac{n k_0 (\overline{x}_n - \mu_0)^2}{n + k_0} \right) 2\sigma^2 \right\}$$

Full normal: Normal-Gamma prior

The posterior of $\mu, \sigma^{-2} | x_1, \ldots, x_n$ is a Normal-Gamma distribution of parameters

$$\alpha_n = \alpha_0 + n/2$$

$$k_n = k_0 + n$$

$$\mu_n = \frac{k_0 \mu_0 + n \bar{x}_n}{k_0 + n}$$

$$\beta_n = \beta_0 + \frac{s_x^2}{2} + \frac{n k_0 (\overline{x}_n - \mu_0)^2}{2(n + k_0)}$$

Full Normal model: marginal prior/posterior

Normal-(inverse) Gamma prior

If (μ, σ^{-2}) has a Normal-Gamma distribution

$$\sigma^{-2} \sim \mathcal{G}(\alpha_0, \text{rate} = \beta_0) \quad \mu|\sigma^2 \sim \mathcal{N}(\mu_0, \sigma^2/k_0)$$

then the marginal distribution of μ is a Student T

$$\mu \sim \mathcal{T}_{2\alpha_0}(\mu_0, \beta_0/(\alpha_0 k_0))$$

$\mu \sim \mathcal{T}_r(m, s^2)$ if its density is proportional to

$$\propto \frac{1}{s(1 + (\mu - m)^2/rs^2)^{(r+1)/2}}.$$

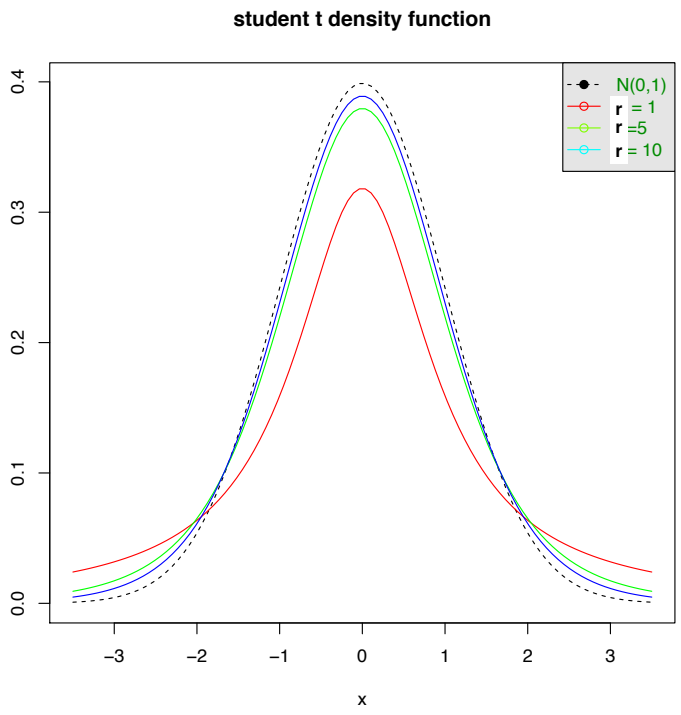
For $m = 0$ and $s = 1$ we recover the standard Student T, $\mathcal{T}_r$.

Student T distribution

Let $T \sim \mathcal{T}_r$.

1. $T^* = sT + m$ is a $\mathcal{T}_r(m, s^2)$, that is m is a position parameter and s a scale parameter.
2. Conversely, if $T^* \sim \mathcal{T}_r(m, s^2)$, then $(T^* - m)/s \sim \mathcal{T}_r$.
3. Since T is symmetric, if $t_{\alpha/2,r}$ is the $\alpha/2$ quantile of T, then

$$\mathbb{P}\left(-t_{\alpha/2,r} < T < t_{\alpha/2,r}\right) = 1-\alpha.$$



Full Normal model: predictive

Predictive distribution - full normal model

In the full normal model the (posterior) predictive distribution for x_{n+1} is

$$x_{n+1}|x_1, \ldots, x_n \sim \mathcal{T}_{2\alpha_n}(\mu_n, \beta_n(k_n+1)/(\alpha_n k_n)).$$

Recall that

$$\alpha_n = \alpha_0 + n/2, \quad k_n = k_0 + n,$$
$$\mu_n = \frac{k_0\mu_0 + n\bar{x}_n}{k_0 + n}, \quad \beta_n = \beta_0 + \frac{s_x^2}{2} + \frac{nk_0(\overline{x}_n - \mu_0)^2}{2(n+k_0)}.$$

Exercise with R

Use the data:

https://www.ceremade.dauphine.fr/~xian/BCS/normaldata

1. Choose prior parameters $(\alpha_0, \beta_0, \mu_0, k_0)$ (use small values for α_0 and β_0... why?).
2. Compute the posterior distribution of μ and plot it.
3. Compute the predictive distribution of x_{new}.
4. Plot the predictive density against the histogram of the data.

Exercise with R

Hint: Instal the LaplacesDemon package and use the function ”dst”, which is the density of a T-student with additional position and scale parameters.

- install.packages(”LaplacesDemon”)
- library(LaplacesDemon)
- use the command dst, use help(dst) to see how does it work
- ... ”dst(x, mu=PIPPO, sigma=PLUTO, nu=PETRONIO, log=FALSE)”

SEE the SECOND NOTEBOOK and the FIRST LAB

Chapter 3.

- decisions
- credible regions
- testing hypotheses
- improper priors

Bayesian learning: decision and loss

Purpose of most inferential studies: to provide the statistician/client with a *decision* $d \in D$
Requires an evaluation criterion/loss function for decisions and estimators.

- When making inference $D = \Theta$, the decision is "select a parameter as your estimate".

Bayesian learning: Bayes estimator

A loss function is $L(\theta, d) : \Theta \times D \to \mathbb{R}_+$, where

- Θ is the parameter space,
- D is a *decision space*.

One has that $L(\theta, d)$ is the loss incurred when θ is the parameter and d is the decision.

- $d \in D$ is an *elementary decision*.
- the statistical decision is a function of the samples, $\delta(x) : \mathcal{X} \to D$.

Bayes estimator

A *Bayes estimate* associated with a prior distribution π and a loss function L is

$$\delta(x) = \arg\min_{\delta} \mathbb{E}[L(\theta, \delta(x))|x] \,.$$

Bayesian learning: Estimation

Examples, assuming $\Theta \subset \mathbb{R}$:

- Using quadratic loss, one gets the posterior mean:

$$\theta_{mean}(x) = \mathbb{E}[\theta|x] = \frac{\int_\Theta \theta f(x|\theta)\pi(\theta)\, d\theta}{\int_\Theta f(x|\theta)\pi(\theta)\, d\theta}$$

- Using absolute value loss, one gets the posterior median:

$$\theta_{med}(x) = \text{median of } \pi(\theta|x)$$

- With no loss, one can use the maximum a posteriori (MAP) estimator:

$$\theta_{MAP}(x) = \arg\max_\theta f(x|\theta)\pi(\theta)$$

!! compare it with the MLE !!

Loss functions (optional)

Principle. Integrate over the space Θ to get the posterior expected loss

$$\begin{aligned}\rho(\pi, \delta) &= \mathbb{E}[L(\theta, \delta(x))|x] \\ &= \int_{\Theta} L(\theta, \delta(x))\pi(\theta|x)\, d\theta,\end{aligned}$$

and minimise in δ to get the Bayes estimator.

Risk function

The loss function can be used to define the *risk function* (frequent risk function)

$$R(\theta, \delta) = \int_{\mathcal{X}} L(\theta, \delta(x)) f(x|\theta) dx$$

Impossible to uniformly minimize (in δ) the loss function. The risk function cannot be used to define an optimal decision (estimator) uniformly in θ.

(optional) .

Note that the *integrated risk* function is

$$
\begin{aligned}
r(\pi,\delta) &= \int_\Theta R(\theta,\delta)\pi(\theta)d\theta \\
&= \int_\Theta \int_{\mathcal{X}} L(\theta,\delta(x))f(x|\theta)dx\pi(\theta)d\theta \\
&\overset{Fubini}{=} \int_{\mathcal{X}} \left(\int_\Theta L(\delta(x),\theta)\pi(\theta|x)d\theta\right) m(x)dx
\end{aligned}
$$

where

$$
\int_\Theta L(\delta(x),\theta)\pi(\theta|x)d\theta = \rho(\pi,\delta(x))
$$

is the *posterior expected loss*.

So that minimizing $\rho(\pi,\delta(x))$ for any given x one minimizes also $r(\pi,\delta)$.

The quadratic & abs. val. loss (optional)

For $\Theta \subset \mathbb{R}$ one can choose

$$L(\theta, d) = (\theta - d)^2$$

The Bayes estimate $\delta(x)$ associated with the prior π and with the quadratic loss is the posterior expectation

$$\delta(x) = \mathbb{E}[\theta|x] = \frac{\int_\Theta \theta f(x|\theta)\pi(\theta)\, d\theta}{\int_\Theta f(x|\theta)\pi(\theta)\, d\theta}.$$

Indeed in this case

$$\rho(\pi, \delta(x)) = \mathbb{E}[|\theta - \delta(x)|^2|x].$$

Alternatives to the quadratic loss:

$$L(\theta, d) =| \theta - d |,$$

The associated Bayes estimate is the median of of $\pi(\theta|x)$.

Bayesian learning: Credible regions

$C_\alpha(x) \subset \Theta$ is a bayesian credible region of level α for θ if

$$Prob\{\vartheta \in C_\alpha(X)|X = x\} = \pi(C_\alpha(x)|x) = 1 - \alpha.$$

The frequentist answer is the confidence region

$$Prob\{\theta_{true} \in C_\alpha(X)|\vartheta = \theta_{true}\} = 1 - \alpha \quad \forall\ \theta_{true}.$$

typically: $\alpha = 0.05, 0.01, \ldots.$

Confidence vs. Credibility

IMPORTANT: the bayesian credible region may be equal to the frequentist confidence interval, but in general it is not. More importantly it has a different interpretation.

- "Given our observed data, there is a 95% probability that the (true) value of the parameter falls within $C_{.05}$" - Bayesians
- "There is a 95% probability that when I compute a C.I. from data of this sort, the true mean will fall within $C_{.05}$." - Frequentists

Taken from

http://jakevdp.github.io/blog/2014/06/12/frequentism-and-bayesianism-3-confidence-credibility/

Bayesian learning: Credible regions

Highest posterior density (HPD) regions are of the form

$$C_\alpha = \{\theta : \pi(\theta|x) > k_\alpha\}$$

where k_α is chosen in such a way that C_α has level α.

Optimality

The HPD regions give the highest probabilities of containing θ for a given volume

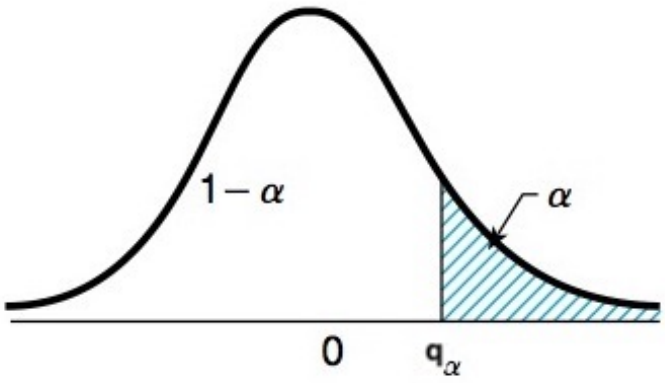
Quantile *(PrRe)*

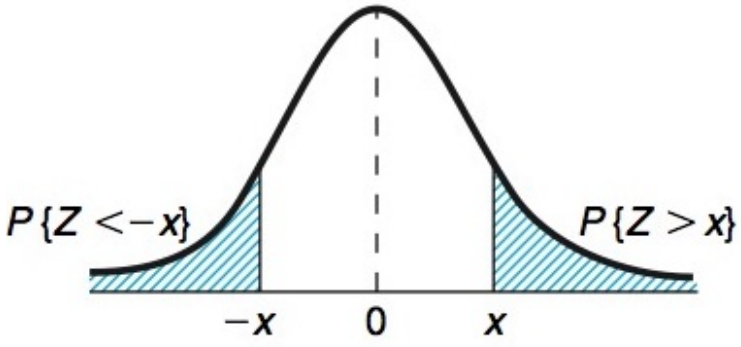
We say that the number q_α is the $100(1-\alpha)$ percentile (or α quantile) of a random variable X if

$$P\{X > q_\alpha\} = \alpha$$

If the distribution of X is symmetric (and continuous)

$$P\{-q_{\alpha/2} \leq X \leq q_{\alpha/2}\} = 1-\alpha.$$





Quantile *(PrRe)*

If the posterior distribution of $\tilde{\theta} := (\theta - c)/b$ is symmetric (for a suitable caentering and normalizing constants b, c), and $q_{\alpha/2}$ is the $\alpha/2$ quantile of $\tilde{\theta}$ (under the posterior distribution) then

$$C_\alpha = (c - q_{\alpha/2} b, c + q_{\alpha/2} b)$$

is a credible interval of level α. Indeed,

$$\begin{aligned}\pi\{\theta \in C_\alpha | x\} &= \pi\Big\{ - q_{\alpha/2} \leq \frac{(\theta - c)}{b} \leq q_{\alpha/2} | x \Big\} \\ &= \pi\{-q_{\alpha/2} \leq \tilde{\theta} \leq q_{\alpha/2} | x\} = 1 - \alpha.\end{aligned}$$

Note that it is is also a HPD.
[Why?]

Full Normal model: intervals

The posterior $\mu, \sigma^{-2}|x_1, \ldots, x_n$ is a Normal-Gamma of parameters

$$\alpha_n = \alpha_0 + n/2, \quad k_n = k_0 + n,$$
$$\mu_n = \frac{k_0\mu_0 + n\bar{x}_n}{k_0 + n}, \quad \beta_n = \beta_0 + \frac{s_x^2}{2} + \frac{nk_0(\overline{x}_n - \mu_0)^2}{2(n + k_0)}.$$

and hence that marginal posterior of μ is $\mathcal{T}_{2\alpha_n}(\mu_n, \beta_n/(\alpha_n k_n))$. Recalling that μ_n is a centering parameter and $\omega_n := \sqrt{\beta_n/(\alpha_n k_n)}$ a scale parameter we get:

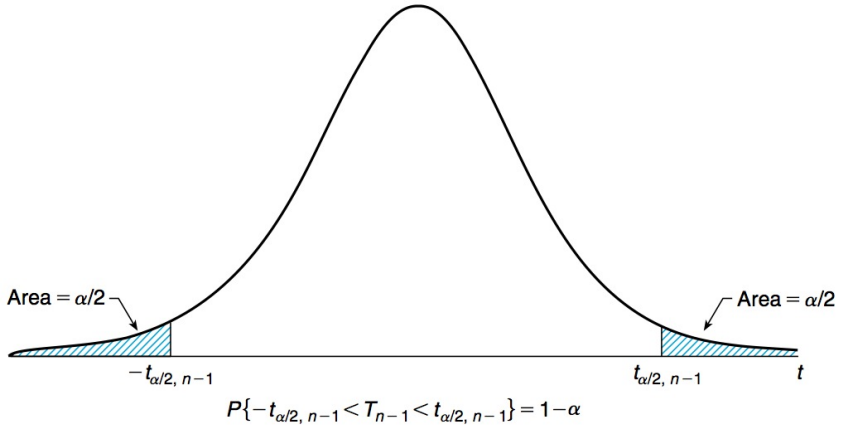
Credible intervals - full normal model

A credible interval at level $\alpha \in (0, 1)$ for μ in the full normal model (normal-inverse gamma prior) is

$$C_\alpha^\pi = \left[\mu_n - t_{\alpha/2,2\alpha_n}\omega_n, \mu_n + t_{\alpha/2,2\alpha_n}\omega_n\right],$$

where $t_{\alpha/2,r}$ is the $\alpha/2$-quantile of a $\mathcal{T}_r$ distribution.

good to know



A two-sided frequentist confidence interval for the parameter μ is

$$\left(\overline{x}_n - t_{\alpha/2,n-1}\frac{s_x}{\sqrt{(n-1)n}},\ \overline{x}_n + t_{\alpha/2,n-1}\frac{s_x}{\sqrt{(n-1)n}}\right)$$

where

$$s_x = \sqrt{\sum_{i=1}^{n}(x_i - \bar{x}_n)^2}.$$

Exercise with normaldata

normaldata

Compute the 95% confidence region for μ using prior parameters $\mu_0 = 0, k_0 = 0.1, a_0 = 0.1, b_0 = 0, 1$. Is it acceptable reporting a significant decrease in the number of larcenies between 1991 and 1995?

Testing hypotheses

The question of the previous exercise:

"Is it acceptable reporting a significant decrease in the number of larcenies between 1991 and 1995?"

can be translated into a statement concerning the mean of the normal distribution: is the mean μ less or equal to zero or not? Or also: do we "prefer"

$$\Theta_0 := \{\mu \leq 0\} \quad \text{or} \quad \Theta_0^c := \{\mu > 0\}?$$

This is a typical problem of testing hypothesis.

Testing hypotheses

Deciding about validity of assumptions or restrictions on the parameter θ from the data. Reesented as

$$H_0 : \theta \in \Theta_0 \quad \text{versus} \quad H_1 : \theta \notin \Theta_0$$

Binary outcome of the decision process: *accept* [coded by 1] or *reject* [coded by 0]

$$\mathcal{D} = \{0, 1\}$$

The $0-1$ *loss*

Rudimentary loss function

$$L(\theta, d) = \begin{cases} 1-d & \text{if } \theta \in \Theta_0 \\ d & \text{otherwise,} \end{cases}$$

Associated Bayes estimate

$$\delta^\pi(x) = \begin{cases} 1 & \text{if } \pi(\theta \in \Theta_0 | x) > \dfrac{1}{2}, \\ 0 & \text{otherwise.} \end{cases}$$

Intuitive structure

Full-normal model

Recall that the posterior of μ is $\mathcal{T}_{2\alpha_n}(\mu_n, \beta_n/(\alpha_n k_n))$. Hence, for a given m_0 and $\Theta_0 = \{\mu \leq m_0\}$ then

$$\begin{aligned}\pi\{\Theta_0|x_1,\ldots,x_n\} &= P\{\mu \leq m_0|x_1,\ldots,x_n\} \\ &= P\{(\mu-\mu_n)/\omega_n \leq (m_0-\mu_n)/\omega_n|x_1,\ldots,x_n\}\end{aligned}$$

Since $(\mu-\mu_n)/\omega_n \sim \mathcal{T}_{2\alpha_n}$ one has

$$\pi\{\Theta_0|x_1,\ldots,x_n\} = F_{t,2\alpha_n}((m_0-\mu_n)/\omega_n)$$

where $F_{t,2\alpha_n}$ is the cdf of a standard Student T with $2\alpha_n$ degree of freedom.

Exercise with normaldata

normaldata

Compute the posterior probabilities $\pi\{\mu \geq 0|x\}$ and $\pi\{\mu > 0|x\}$ using prior parameters $\mu_0 = 0, k_0 = 0.1, a_0 = 0.1, b_0 = 0, 1$. Is it acceptable reporting a significant decrease in the number of larcenies between 1991 and 1995?

Test and Bayes factor

Let us rewrite the prior as

$$\pi(\theta) = w_1 \pi_1(\theta) + w_0 \pi_0(\theta)$$

where

$$w_1 = \pi(\theta \in \Theta_1) = \int_{\Theta_1} \pi(\theta) d\theta \quad w_0 = \pi(\theta \in \Theta_0) = \int_{\Theta_0} \pi(\theta) d\theta$$

and for $i = 0, 1$

$$\pi_i(\theta) = \begin{cases} \frac{\pi(\theta)}{\int_{\Theta_i} \pi(\theta) d\theta} & \text{if } \theta \in \Theta_i \\ 0 & \text{if } \theta \notin \Theta_i \end{cases}$$

Test and Bayes factor

.. then

$$\pi\{\Theta_i|x\} = \frac{w_i \int_{\Theta_i} f(x|\theta_i)\pi_i(\theta_i)d\theta_i}{w_0 \int_{\Theta_0} f(x|\theta_0)\pi_0(\theta_0)d\theta_0 + w_1 \int_{\Theta 1} f(x|\theta_1)\pi_1(\theta_1)d\theta_1}$$
$$= \frac{w_i m_i(x)}{w_0 m_0(x) + w_1 m_1(x)}$$

where $m_i(x)$ denotes the marginal density of the data under hypothesis $i = 0, 1$. Since the posterior probabilities are sensitive to the priors w_0 and w_1 one can use the Bayes factor...

Bayes factor

The **Bayes factor** is defined as

$$B_{10}^{\pi} = \frac{\pi(\theta \in \Theta_1 | x)}{\pi(\theta \in \Theta_0 | x)} \frac{\pi(\theta \in \Theta_0)}{\pi(\theta \in \Theta_1)} = \frac{m_1(x)}{m_0(x)}$$

Note that B_{10} does not depend on w_1, w_0.

If $\mathcal{M}_1$ and $\mathcal{M}_0$ are the (sub)-models corresponding to Θ_0 and Θ_1, these models can be compared using

$$B_{10}^{\pi} = \frac{m_1(x)}{m_0(x)}$$

One also has

$$\pi\{\Theta_0 | x\} = \left[1 + \frac{w_1}{w_0 BF_{10}}\right]^{-1}$$

Jeffreys' scale

1. if $\log_{10}(B_{10}^{\pi}) < 0$ the is *no evidence* against H_0,
2. if $\log_{10}(B_{10}^{\pi})$ varies between 0 and 0.5, the evidence against H_0 is *poor*,
3. if it is between 0.5 and 1, it is is *substantial*,
4. if it is between 1 and 2, it is *strong*, and
5. if it is above 2 it is *decisive*.

Point null difficulties

If π absolutely continuous then

$$\pi(\theta = \theta_0) = 0 \ldots$$

$$\boxed{\textbf{How can we test } H_0 : \theta = \theta_0\textbf{?!}}$$

Clearly if $\pi(\theta = \theta_0) = 0$, also $\pi(\theta = \theta_0 | x) = 0$...

New prior for new hypothesis

Testing point null difficulties requires a modification of the prior distribution so that

$$\pi(\Theta_0) > 0 \quad \text{and} \quad \pi(\Theta_1) > 0$$

E.g., when $\Theta_0 = \{\theta_0\}$, π_0 *is Dirac mass at* θ_0. That is

$$\pi(\theta) = (1-\rho)\pi_1(\theta) + \rho\delta_{\theta_0}(\theta)$$

(here $\rho = w_0$ of the previous slides, e.g. one can take $\rho = 0.5$). Recall that

$$\delta_{\theta_0}(A) = \begin{cases} 1 & \text{if } \theta_0 \in A \\ 0 & \text{if } \theta_0 \notin A \end{cases}$$

Posteriors with Dirac masses

If $H_0 : \ \theta = \theta_0 \ (= \Theta_0)$, and

$$\pi(\theta) = \rho\delta_{\theta_0}(\theta) + (1-\rho)\pi_1(\theta)$$

then

$$\pi(\Theta_0|x) = \frac{f(x|\theta_0)\rho}{\int f(x|\theta)\pi(\theta)\,d\theta} = \frac{f(x|\theta_0)\rho}{f(x|\theta_0)\rho + (1-\rho)m_1(x)}$$

with

$$m_1(x) = \int_{\Theta_1} f(x|\theta)\pi_1(\theta)\,d\theta.$$

Laplace's prior and improper prior

Principle of *Insufficient Reason* (Laplace)

$$\Theta = \{\theta_1, \cdots, \theta_p\} \qquad \pi(\theta_i) = 1/p$$

(no hyperparameters to choose)
Extension to continuous spaces

$$\pi(\theta) \propto 1$$

Lebesgue measure

!!! on $\mathbb{R}$ it is not a probability measure !!!

Improper prior distribution

Extension from a prior distribution to a prior σ-finite measure π such that

$$\int_{\Theta} \pi(\theta)\, d\theta = +\infty$$

Formal extension: π cannot be interpreted as a probability any longer

Validation

Extension of the posterior distribution $\pi(\theta|x)$ associated with an improper prior π given by Bayes's formula

$$\pi(\theta|x) = \frac{f(x|\theta)\pi(\theta)}{\int_\Theta f(x|\theta)\pi(\theta)\,d\theta},$$

when

$$\int_\Theta f(x|\theta)\pi(\theta)\,d\theta < +\infty$$

Full normal model with improper prior

Under the improper prior

$$\pi(\mu, \sigma^2) = 1/\sigma^2,$$

(formally $\alpha_0 = -0.5$, $\beta_0 = 0$, $k_0 = 0$,$\mu_0 = 0$) posterior distribution of (μ, σ)

$$\begin{aligned} \mu|\sigma, \bar{x}, s_x^2 &\sim \mathcal{N}\left(\bar{x}, \frac{\sigma^2}{n}\right), \\ \sigma^2|\bar{x}, s_x^2 &\sim \mathcal{IG}\left(\frac{n-1}{2}, \frac{s_x^2}{2}\right). \end{aligned}$$

Then

$$\pi(\mu|\bar{x}, s_x^2) \propto \left[s_x^2 + n(\bar{x} - \mu)^2\right]^{-n/2}$$

that is a $\mathcal{T}_{n-1}(\bar{x}, s_x^2)$ distribution

Normal credible interval

Derived credible interval on μ

$$[\bar{x} - t_{\alpha/2,n-1} s_x / \sqrt{n(n-1)}, \bar{x} + t_{\alpha/2,n-1} s_x / \sqrt{n(n-1)}]$$

normaldata

Corresponding 95% confidence region for μ

$$[-0.070, -0.013,]$$

Since 0 does not belong to this interval, reporting a significant decrease in the number of larcenies between 1991 and 1995 is acceptable

Exponential families *(optional)*

Exponential family

Let $T : \mathcal{X} \to \mathbb{R}^k$ and $R : \Theta \to \mathbb{R}^k$. The family of distributions

$$f(x|\theta) = C(\theta)h(x)\exp\{R(\theta) \cdot T(x)\}$$

is called an *exponential family of dimension k*. When $\Theta \subset \mathbb{R}^k$, $\mathcal{X} \subset \mathbb{R}^k$ and

$$f(x|\theta) = h(x)\exp\{\theta \cdot x - \Psi(\theta)\},$$

the family is said to be *natural*.

$\Psi(\theta) = \log(1/C(\theta))$ and $C(\theta)$ is the normalizing constant:

$$C(\theta)^{-1} = \begin{cases} \int_{\mathcal{X}} h(x)\exp\{R(\theta) \cdot T(x)\}dx & \text{continuous case} \\ \sum_{x \in \mathcal{X}} h(x)\exp\{R(\theta) \cdot T(x)\} & \text{discrete case} \end{cases}$$

Analytical properties of exponential families *(optional)*

- Common enough structure (normal, Poisson, &tc...)
- Allow for conjugate priors

$$\pi(\theta|\mu,\lambda) = K(\mu,\lambda)\, e^{\theta\cdot\mu-\lambda\Psi(\theta)} \qquad \lambda > 0$$

(optional)

In the canonical form

$$\pi(\theta|\mu,\lambda) = K(\mu,\lambda)\, e^{\theta\cdot\mu-\lambda\Psi(\theta)}$$

$$f(x|\theta) = h(x)e^{\theta\cdot x-\Psi(\theta)},$$

and hence

$$f(x|\theta)\pi(\theta|\mu,\lambda) \propto e^{\theta\cdot x-\Psi(\theta)}e^{\theta\cdot\mu-\lambda\Psi(\theta)} \propto e^{\theta\cdot(x+\mu)-(\lambda+1)\Psi(\theta)}$$

The parameters are updated as follows (given one boservation!!)

$$\mu \to (x+\mu) \quad \lambda \to (\lambda+1)$$

Standard exponential families

$f(x\|\theta)$	$\pi(\theta)$	$\pi(\theta\|x)$
Normal $\mathcal{N}(\theta, \sigma^2)$	Normal $\mathcal{N}(\mu, \tau^2)$	$\mathcal{N}(\rho(\sigma^2\mu + \tau^2 x), \rho\sigma^2\tau^2)$ $\rho^{-1} = \sigma^2 + \tau^2$
Poisson $\mathcal{P}(\theta)$	Gamma $\mathcal{G}(\alpha, \eta)$	$\mathcal{G}(\alpha + x, \eta + 1)$
Gamma $\mathcal{G}(\nu, \theta)$	Gamma $\mathcal{G}(\alpha, \eta)$	$\mathcal{G}(\alpha + \nu, \eta + x)$
Binomial $\mathcal{B}(n, \theta)$	Beta $\mathcal{B}e(\alpha, \eta)$	$\mathcal{B}e(\alpha + x, \eta + n - x)$

More...

$f(x\|\theta)$	$\pi(\theta)$	$\pi(\theta\|x)$
Negative Binomial $\mathcal{N}eg(m,\theta)$	Beta $\mathcal{B}e(\alpha,\eta)$	$\mathcal{B}e(\alpha+m,\eta+x)$
Multinomial $\mathcal{M}_k(\theta_1,\ldots,\theta_k)$	Dirichlet $\mathcal{D}(\alpha_1,\ldots,\alpha_k)$	$\mathcal{D}(\alpha_1+x_1,\ldots,\alpha_k+x_k)$
Normal $\mathcal{N}(\mu,1/\theta)$	Gamma $\mathcal{G}a(\alpha,\eta)$	$\mathcal{G}(\alpha+0.5,\eta+(\mu-x)^2/2)$